

MovieLens Feature Engineering and Exploratory Data Analysis Report

1. Introduction

This report presents the Stage 1 deliverables of the internship task, which involved performing Feature Engineering and Exploratory Data Analysis (EDA) on the MovieLens dataset. The dataset includes information about movies, user ratings, tags, and related metadata. The main aim of this stage was to clean the dataset, create new analytical features, and visualise key patterns that illustrate user preferences and movie characteristics.

2. Data Cleaning Summary

Data preparation was conducted to ensure that the dataset was accurate, consistent, and complete before analysis. The following steps were implemented during cleaning:

- Missing tags were replaced with 'No Tag' to avoid null-related errors.
- Duplicate records were checked and removed.
- Rows with missing critical identifiers (tmdbId) were dropped when there were few to avoid mismatches.
- Timestamps were converted into readable date formats (year and month).

3. Feature Engineering (with reasons)

Feature engineering was conducted to create new columns that provide deeper insight into movie popularity, user engagement, and temporal trends. Each feature below includes the reason it was created and how it helps the analysis.

- **Release Year (release_year):** Extracted from the movie title. Enables temporal grouping (by year or decade) to analyse rating trends over time, detect era-related preferences, and control for age in comparative analyses.
- **Number of Genres (num_genres):** Counts genres per movie. Captures the thematic diversity of a movie. Multi-genre films may appeal to broader audiences; this helps test whether genre diversity correlates with higher ratings.
- **Average Rating (avg_rating):** Mean rating per movie. This is the Core measure of audience sentiment and perceived quality — used to rank movies and compare performance across groups (genre, decade, age group).

- **Number of Ratings per Movie (num_ratings):** Count of ratings per movie (derived from aggregation). This is used to measure popularity and reliability — movies with many ratings provide more stable average estimates and identify mainstream titles vs niche ones.
- **User Rating Frequency (user_rating_freq):** Number of ratings each user submitted. It identifies 'power users' and allows checks for rater bias; it helps determine whether a small group of active users skews ratings.
- **Movie Age (movie_age):** Computed as $2025 - \text{release_year}$. Quantifies recency/legacy effect. Used to analyse nostalgia or recency bias (e.g., whether newer movies receive systematically different ratings).
- **Number of Tags (num_tags):** Count of tags per movie. It serves as a proxy for engagement and discussion frequency — highly tagged movies are often culturally salient or controversial, which can correlate with popularity or distinctive rating patterns.
- **Year and Month (year, month, month_year):** Extracted from rating timestamps. Allows temporal analysis of when ratings were given (seasonality, recent activity) and supports time-series visualisations or trend detection.

4. Exploratory Data Analysis (EDA)

The EDA phase focused on understanding rating behaviour, temporal trends, and relationships between movie characteristics. Visualisations were created using Matplotlib and Seaborn to highlight key patterns and trends. Each figure below represents a major analytical question explored.

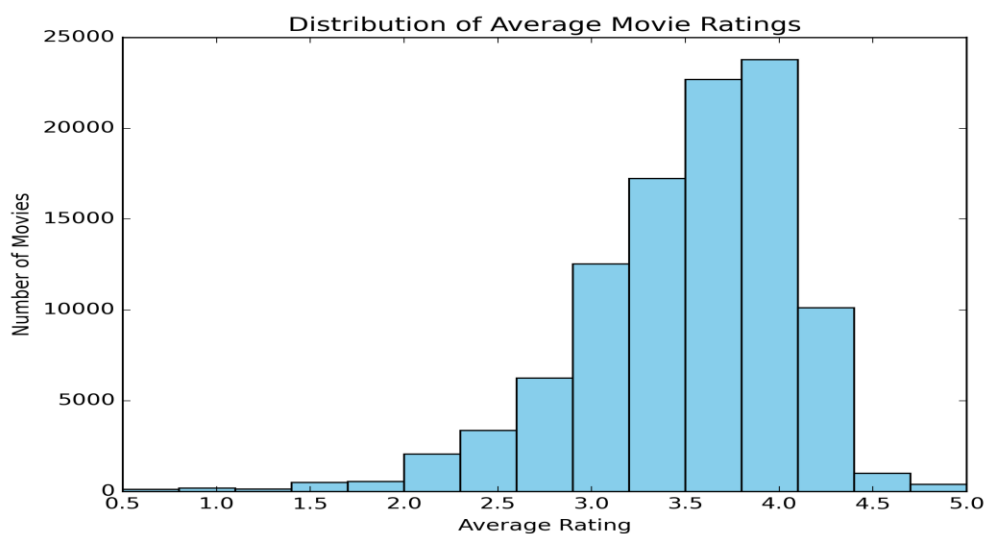


Figure 1: Distribution of Average Ratings

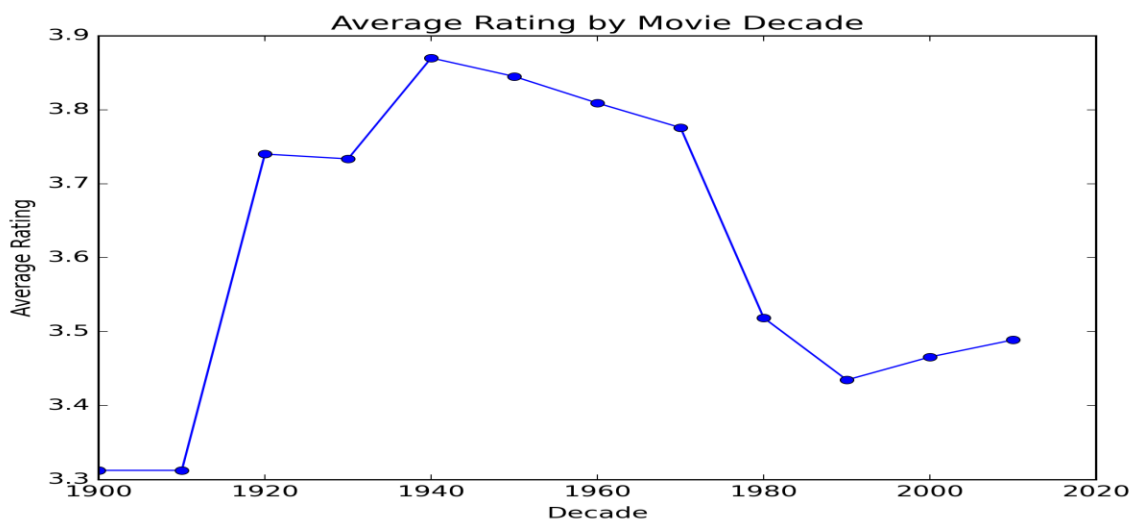


Figure 2: Average Rating by Movie Decade

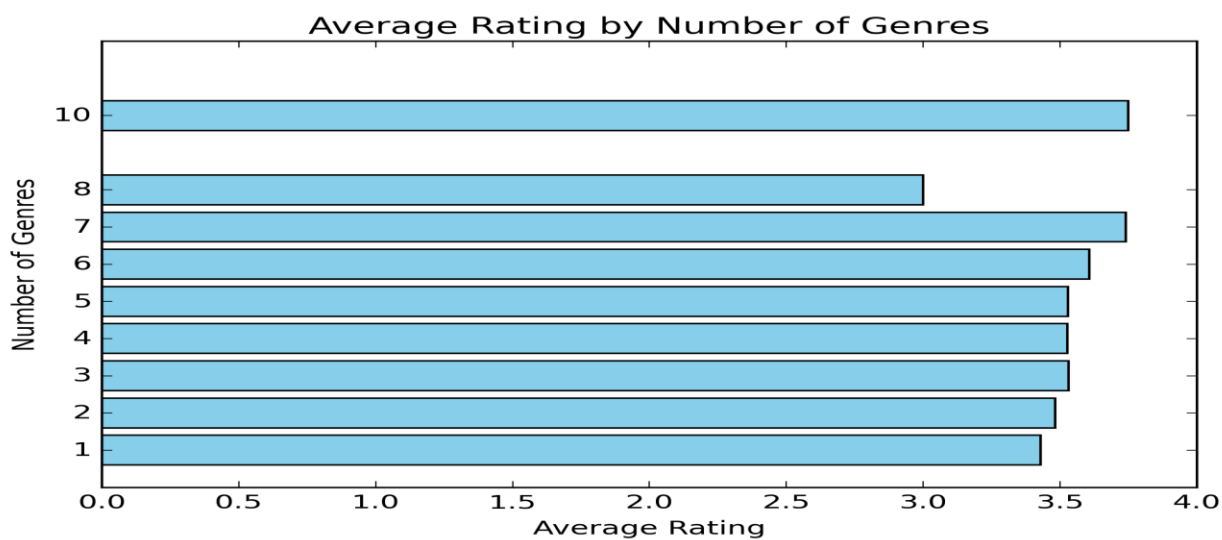


Figure 3: Average Rating by Number of Genres

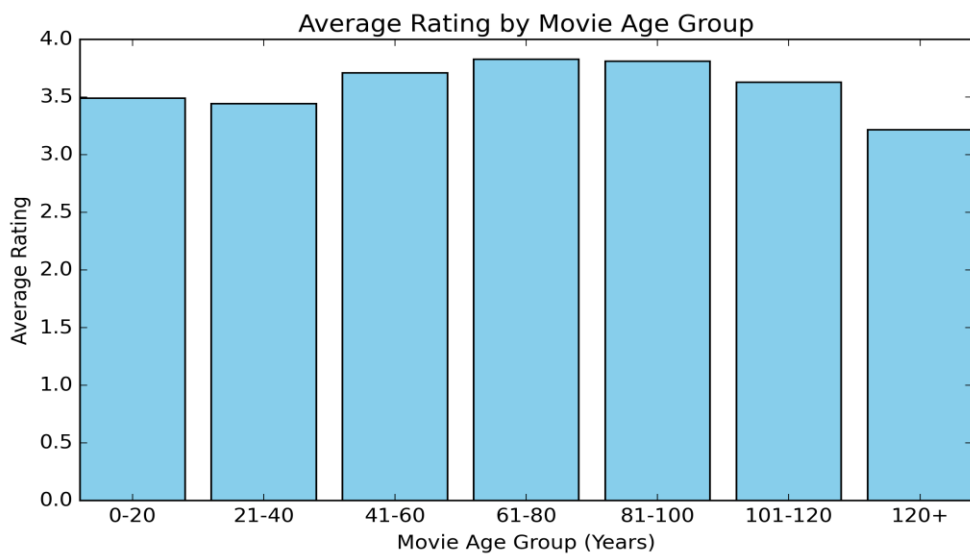


Figure 4: Average Rating by Movie Age Group

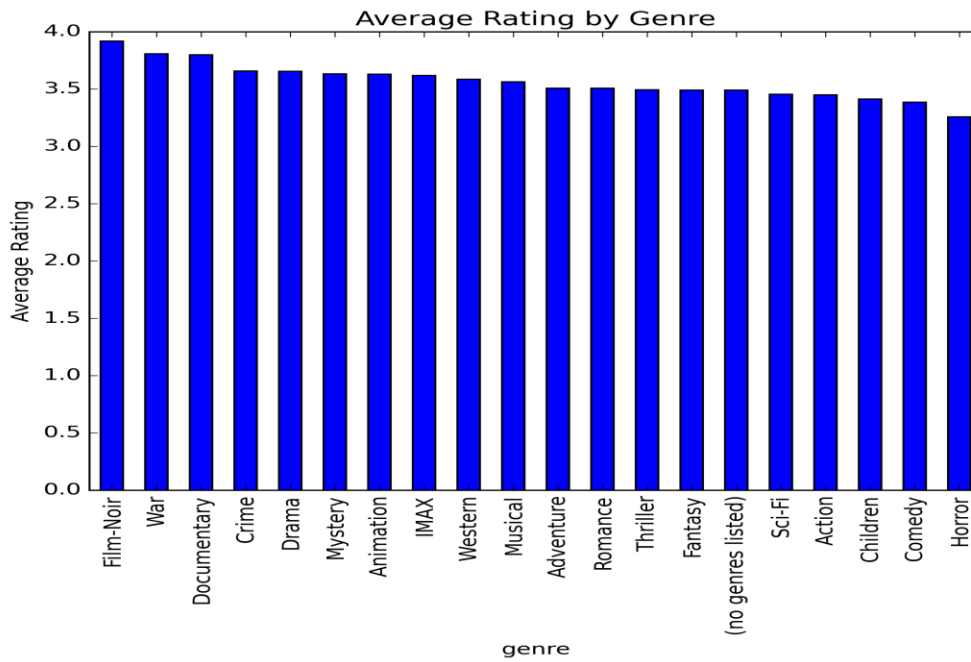


Figure 5: Average Rating by Genres

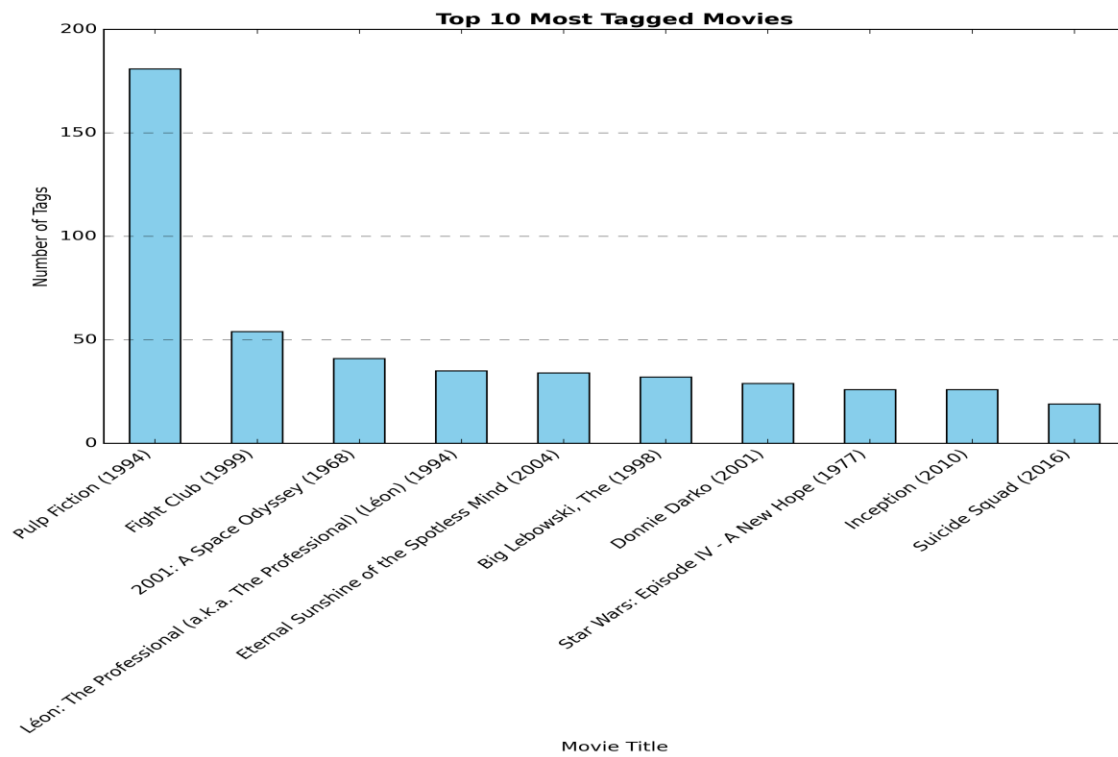


Figure 6: Top 10 Most Tagged Movies

5. Insights and Recommendations

1. Distribution of Average Ratings

Most movies sit comfortably between 3.0 and 4.0 stars, showing that viewers usually rate films they already expect to enjoy. Low ratings are rare, hinting that people either skip bad movies or rate generously. Only a few exceptional titles break the 4.5 mark — proof that true favourites are rare gems.

2. Average Rating by Movie Decade

Ratings peak in the 1940s, when fewer films were made but many became timeless.

As the decades roll on, averages dip slightly — more movies, more mediocrity. Recent years show a small rebound, maybe thanks to streaming and better production quality. It's a reminder that nostalgia and quality shape how we remember films.

3. Average Rating by Number of Genres

Movies with more genres tend to score higher, suggesting that variety keeps audiences interested. Single-genre films can feel predictable, while multi-genre ones seem richer and more creative. The difference is small but meaningful — balance and creativity still matter.

4. Average Rating by Movie Age Group

Films aged 61–100 years shine the brightest, proving that true classics never fade. Newer movies score a bit lower, likely due to high expectations and tougher audiences.

The oldest films struggle most, but they still carry historical charm. Good stories clearly stand the test of time.

5. Average Rating by Genre

Genres like Film-Noir, War, and Documentary top the list for their depth and realism. Meanwhile, Comedy and Horror divide opinion — what makes one person laugh or scream might not work for another. It's a reminder that film tastes personal, not universal.

6. Top 10 Most Tagged Movies

Cult classics like *Pulp Fiction* and *Fight Club* lead the list, showing how certain stories stay alive through fan discussions. Older hits like *2001: A*

Space Odyssey and *Star Wars* still spark engagement decades later. A few films dominate attention, while many quietly hold their place — a perfect example of how cultural impact outlasts time.

Recommendations

The engineered features and insights from this analysis can serve as a foundation for a personalised movie recommendation system.

By combining both user behaviour (through ratings and engagement) and movie characteristics (like age, genre, and popularity), recommendations can be made smarter and more relevant.

Key takeaways:

- **Time-based suggestions:** Use `movie_age` and `release_year` to balance nostalgia with recency bias.
- **Genre diversity:** Leverage `num_genres` and genre ratings to recommend multi-genre, high-rated films.
- **Engagement signals:** Incorporate `num_tags` and `num_ratings` to highlight movies that generate strong audience discussion.
- **User personalisation:** Adjust recommendations for power users and casual viewers to match engagement levels.

Conclusion

This analysis demonstrates how data-driven insights can deepen our understanding of audience behaviour and movie preferences.

From the clear rating trends across decades to the power of genre diversity and user engagement, each feature tells part of the story behind what makes a movie memorable. The findings not only reveal meaningful viewing patterns but also provide a strong base for building intelligent, hybrid recommendation systems that can enhance user satisfaction and discovery on modern streaming platforms.